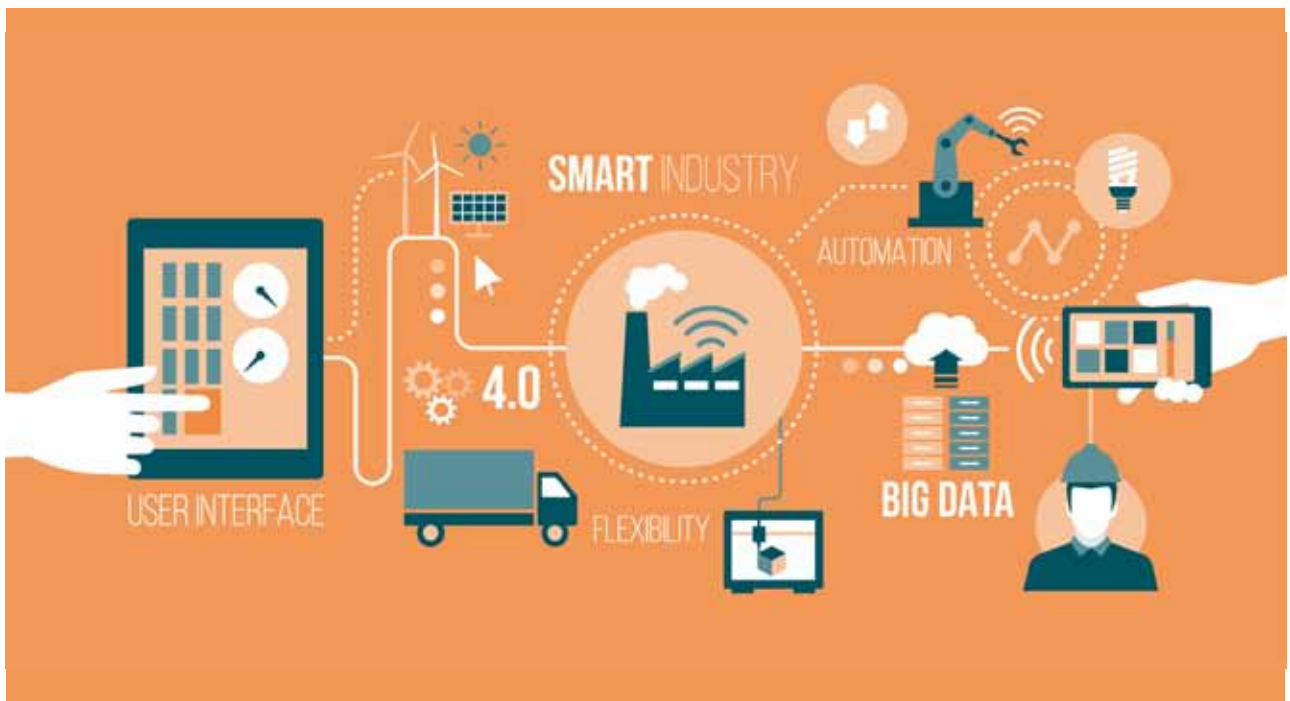


Industrial IoT and Open Source Apps: **Shaping the Future of Manufacturing**

The Industrial Internet of Things (IIoT) is reshaping industries and driving efficiencies through the integration of physical and digital systems. Open source applications are a vital component of this transformation, offering cost-effective, customisable, and secure solutions that enable organisations to harness the power of IIoT.



The Industrial Internet of Things (IIoT) is revolutionising the manufacturing landscape, ushering in an era of increased automation, efficiency, and data-driven decision-making. At the heart of this transformation lies the seamless integration of physical machines with digital technologies, resulting in smart factories and industrial processes. One critical component of this evolution is the utilisation of open source applications in the realm of IIoT.

The Industrial IoT revolution

The concept of IoT refers to the interconnection of everyday objects and devices to the internet, enabling them to send and receive data. In the industrial context, this translates to the convergence of operational technology (OT) and information technology (IT), leading to the creation of Industrial IoT. This fusion empowers organisations to collect real-time data from sensors, devices, and machines, which is then processed, analysed, and leveraged

to enhance productivity, reduce downtime, optimise resources, and drive innovation.

Industrial IoT offers a multitude of benefits to various industries, such as manufacturing, energy, healthcare, transportation, and agriculture. For example, in manufacturing, IIoT allows for predictive maintenance, where sensors on machines detect anomalies and wear-and-tear. This pre-emptively schedules maintenance, prevents costly